

Fungal sporocarp productivity and diversity shaped by weather conditions in *Pinus uncinata* stands

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ABSTRACT

Pinus uncinata is a species frequently growing in the tree line limit of southwestern European subalpine ecosystems, hosting fungal communities that provide relevant ecosystem services. Despite the high importance of these forests for conservation, limited prior research has analysed the sporocarp fruiting patterns. This study aims at filling this gap in knowledge by describing interannual changes in sporocarp productivity, diversity, and community composition over 5 years and identifying possible relationships with the main weather and soil physicochemical characteristics. The studied *P. uncinata* stands had an average annual total sporocarp productivity of 21.60 kg ha⁻¹yr⁻¹ in dry weight, hosting a total of 255 species with productivity-dominant ectomycorrhizal families such as Russulaceae and Tricholomataceae that occurred mostly from mid-August to end September. In contrast, saprotrophs showed more stable productivity values from mid-August to end November with average annual sporocarp productivity of 1.76 kg ha⁻¹yr⁻¹ in dry weight but exhibited higher fungal richness values from the start of October until the end of November. Fungal productivity and species richness were mostly influenced by weather conditions while soil physicochemical properties had implications in shaping the sporocarp community composition. This study quantifies the interannual fungal fruiting pattern of the understudied fungal communities and underlines the importance of climatic variables in shaping fungal fruiting patterns within a climate change context. In addition, this work is also of special relevance since *P. uncinata* ecosystems represent the tree line limit, which makes this species and its associated fungal communities especially sensitive to the potential impacts of global warming.

1. Introduction

Fungi are important ecological components in forest ecosystems, playing a significant role in nutrient cycling and organic matter decomposition (Adnan et al., 2022). Saprotrophic fungi, for instance, break down dead plant and animal organic matter into simpler compounds such as carbon dioxide, water, and nutrients, thereby facilitating their reuse by other organisms in the ecosystem (Bahram and Netherway, 2022). Another group of fungi that is equally essential are the ectomycorrhizal fungi (ECM). These fungi are involved in shaping the structure, aeration, and porosity of the soil in addition to enhancing the water acquisition by trees and consequently improving drought tolerance (Clemmensen et al., 2013; Querejeta, 2017). In addition, ECM fungi appear in different forest tree species from the first life stages,

being fundamental for seedling establishment, providing resistance to root pathogens and nutrient uptake (Smith and Read 2010). One clear example of this close association is Pinaceae, being one of the oldest extant plant families with ectomycorrhizal symbiosis (Hibbett and Matheny, 2009; Tedersoo et al., 2010) and hosts fungal communities that play a key role in *Pinus* forests ecosystems biodiversity and ecosystem services.

Previous research has shown that environmental factors such as weather conditions (e.g., rainfall and temperature) and soil physicochemical properties are key components driving fungal fruiting in pine stands (Alday et al., 2017; Collado et al., 2021; Karavani et al., 2018; Morera et al., 2021, 2022). These factors affect the fruiting phenology and entail interannual changes in fungal occurrence, with differences between ECM and saprotrophic functional groups (Kausarud et al.,

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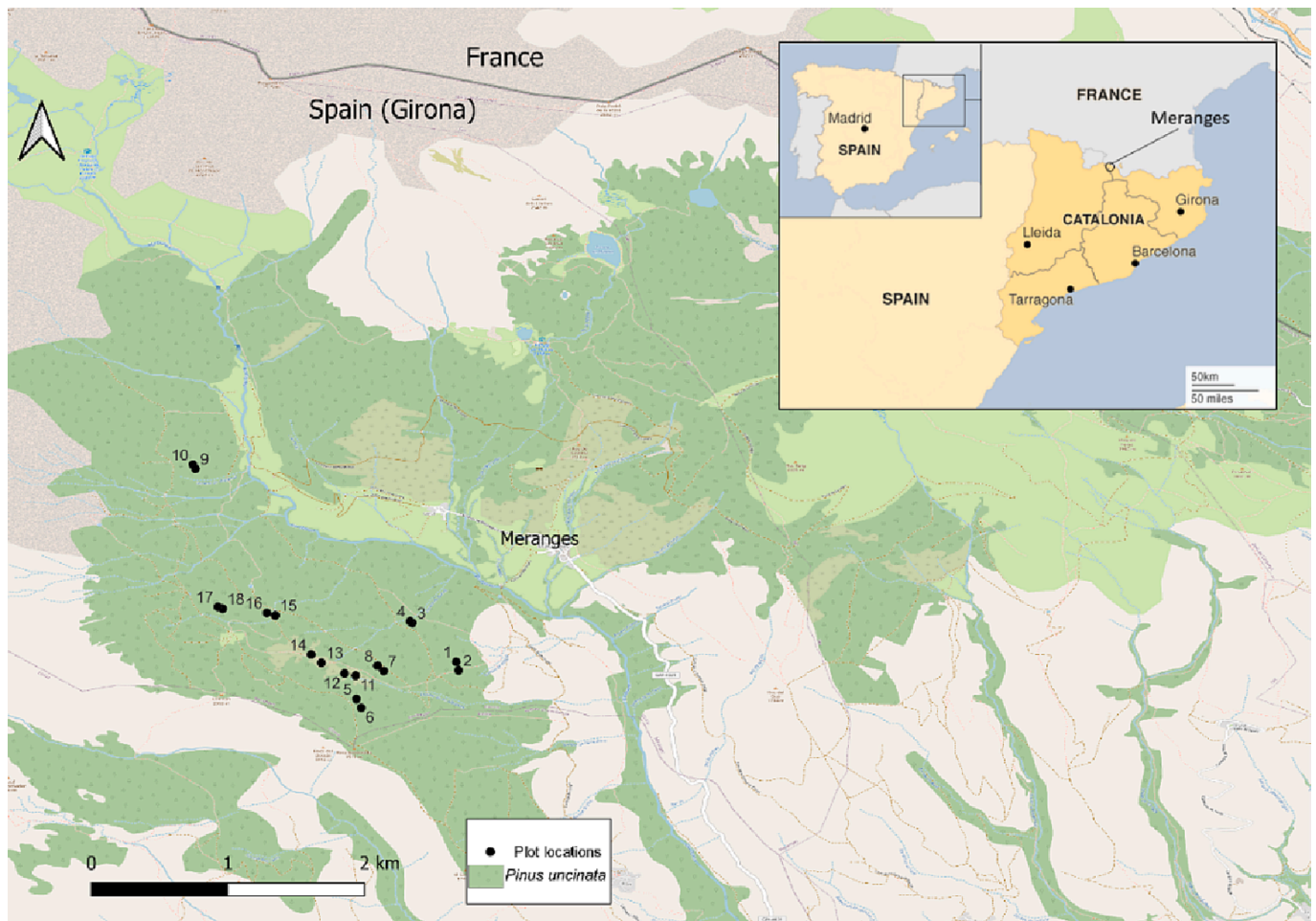


Fig. 1. Map of the study area indicating the studied plots from 2015 to 2019 with dots.

2012; Morera et al., 2022; Ponce et al., 2022). Although it is known that mushrooms are an important non-wood forest product of pine stands (Palahí et al., 2009) and previous research has focused on some pine species (Collado et al., 2021; Palahí et al., 2009) including *P. uncinata* (Solly et al., 2017), important knowledge on sporocarp productivity, diversity and species composition of fungal communities thriving in *Pinus uncinata* forest ecosystems is still missing.

P. uncinata (mountain pine) is a tree species typical of subalpine ecosystems growing in an altitudinal range of 1000 to 2300 m.a.s.l in western Europe, and often representing the tree line limit of altitudinal forest ecosystem distribution (Heuertz et al., 2010). Moreover, since disturbance regimes involving drought and temperature stress can result in an increase in tree species mortality (Anderegg et al., 2013), the rise of temperatures under future climate change scenarios could force the migration of this pine species toward higher altitudes, making these ecosystems and their associated fungal communities especially vulnerable in the face of climate change. It is therefore crucial to have a deeper understanding of the fungal communities thriving in this ecosystem in terms of species composition, sporocarps diversity and productivity, and how they are shaped by different environmental drivers. This can contribute to increasing the knowledge of *P. uncinata* ecosystem functioning and provisioning, remarking the high conservation value of these important areas.

In this study, we aim at expanding the knowledge on fungal fruiting on *P. uncinata* stands by analysing the fungal sporocarp productivity, diversity and community composition in such forest ecosystems. Namely, we aim at: 1) examine the interannual changes in sporocarp productivity, diversity, and community composition of *P. uncinata*

forests, 2) analyse how sporocarp fruiting is driven by different environmental variables such as weather conditions and soil physicochemical properties, and 3) assess the protection status of fungal species according to the IUCN Red List.

2. Material and methods

2.1. Study area

The study area is situated in the Meranges village forest domain (42° 26' 47.705" E, 1° 45' 29.003" N) (Fig. 1) at an altitude of 1539 m.a.s.l. and covering 3423.71 ha in the Catalan Pyrenees (Girona, Spain). The climate is continental with moderate rainfall, low environmental moisture content, high solar irradiance rate and frequent thermal inversion. Winters are cold with mean minimum temperatures of −6.4 °C and warm summers with a mean maximum temperature of 23.7 °C. The mean annual rainfall is 1039 mm, being highly variable between years and as rain from April to October and snowfall from October until March. The stands are composed of a dominating *Pinus uncinata* plantation that ranges from 48 to 120 years with the occasional presence of *P. sylvestris* trees at the lowest elevations. The understorey of some plots is dominated by the presence of *Vaccinium myrtillus*. Moreover, the underlying geology of the studied stands was originated during the Silurian and have a mixture of granite and slate. The soils are deep but acidic with pH ranging from 4.7 to 5.3 (Boixadera and Poch, 2008).

Table 1
Summary of the plots descriptive data.

	Min.	Mean	Median	Max.
Altitude (m.a.s.l.)	1648	1833	1883	1933
Plots slope (%)	7	16	15	34
Stand basal area (m ² ha ⁻¹)	18.41	35.95	37.26	49.46
Stand density (trees ha ⁻¹)	31	106	68	308
pH	4.7	5	5	5.3
Soil organic matter (OM) (%)	3.09	5.60	4.94	8.49

Table 2
Summary of the modelling variables per plot and year from 2015 to 2019.

	Min.	Mean	Median	Max.
Total productivity (dry biomass in kg dw ha ⁻¹)	5.45	21.60 ± 13.09	19.79	52.72
ECM productivity (dry biomass in kg dw ha ⁻¹)	4.63	19.84 ± 12.69	17.32	49.68
Saprotrophic productivity (dry biomass in kg dw ha ⁻¹)	0.06	1.76 ± 1.79	1.12	7.05
Total species richness	7	20 ± 7	21	32
ECM species richness	3	11 ± 5	11	19
Saprotrophic species richness	2	9 ± 4	9	17
Total species evenness	0.39	0.61 ± 0.14	0.62	0.80
ECM species evenness	0.19	0.54 ± 0.22	0.57	0.85
Saprotrophic species evenness	0.10	0.47 ± 0.20	0.47	0.74
July rainfall (mm)	82.18	84.55 ± 1.4	85.01	86.53
August rainfall (mm)	73.06	77 ± 2.31	77.80	80.32
September rainfall (mm)	79.11	82.11 ± 1.73	82.80	84.50
October rainfall (mm)	118.12	122.69 ± 2.66	123.62	126.34
November rainfall (mm)	95.85	101.38 ± 3.21	102.28	105.96
1st 50 mm (Week)	31	34 ± 2	34	35
1st 50 mm (DOY)	222	237 ± 10	243	247
75% annual rainfall (Week)	35	38.4 ± 3	38	42
75% annual rainfall (DOY)	243	270 ± 20	272	296
Autumn rainfall (mm)	96.76	256.50 ± 103.33	224.57	439.80
Number of rainy days	14	27 ± 7	31	34
July min. temperature (°C)	10.81	11.03 ± 0.16	10.99	11.3
July mean temperature (°C)	17.37	18.13 ± 0.55	17.96	19.08
July max. temperature (°C)	24.14	22.74 ± 0.80	22.49	24.14
August min. temperature (°C)	9.25	10.33 ± 0.48	10.37	11.05
August mean temperature (°C)	15.14	17.28 ± 1.01	17.51	19.18
August max. temperature (°C)	18.97	21.79 ± 1.38	22.09	24.53
September min. temperature (°C)	4.41	6.71 ± 1.59	7.21	9.38
September mean temperature (°C)	10.10	13.00 ± 1.67	13.38	15.96
September max. temperature (°C)	13.47	17.09 ± 1.82	17.38	20.24
October min. temperature (°C)	2.77	4.12 ± 0.69	4.23	4.99
October mean temperature (°C)	7.69	9.73 ± 1.38	9.35	12.48
October max. temperature (°C)	10.69	13.38 ± 1.87	12.87	17.49
November min. temperature (°C)	-3.06	-1.08 ± 1.51	-1.33	1.53
November mean temperature (°C)	1.18	3.88 ± 1.75	3.38	7.59
November max. temperature (°C)	3.93	7.10 ± 2.02	6.47	11.53
Cold 1st (Week)	36	39 ± 2	38	43
Cold 1st (DOY)	252	273 ± 18	262	303
Frost 1st (Week)	37	42 ± 2	42	45
Frost 1st (DOY)	258	292 ± 16	295	309

Abbreviations: ECM: ectomycorrhizal; Week: week of the year; DOY: day of the year. Variables explanation is provided in Table A. 1 in Appendix A.

2.2. Sampling design and monitoring

To monitor fungal productivity, diversity, and community composition, 18 permanent sampling plots of 100 m² (10 × 10 m each) were established in 2015 along an altitudinal range from 1648 to 1933 m.a.s.l and slopes ranging from 7 to 34%. The basal area of the studied plots ranged from 18.41 to 49.46 m² ha⁻¹ and the number of trees per plot was from 31 to 308 trees ha⁻¹ (Table 1). Moreover, from 2015 to 2019 each plot was weekly surveyed, from June until the end of the fruiting season marked by frost temperatures, to collect all epigeous sporocarps of ECM and saprotrophic macrofungi. Species forming microscopic or hypogeous sporocarps in addition to fungi fruiting directly on woody debris, fresh dung, or living plants were not considered. Fungal yields were brought to the laboratory for identification at the species level, when possible, using Leica MZ125 stereomicroscope and Leica Q500MC microscope and the necessary reagents, following Courtecuisse and Duhem (2011); Bon (2005); Breitenbach and Kränzlin (1991–2005); Moser (1983); and the trophic group i.e., ECM or saprotrophs was assigned following Agerer (2006), Tedersoo et al. (2014) and Pölme et al. (2020). The nomenclature of the species was done using ‘CABI-Bioscience Database of Fungal Names’, updated to July 2022. Sporocarp species richness was measured as the total number of species collected per plot and year. After identification, sporocarps were counted and dry biomass was measured after drying in air-vented ovens at 35 °C as described by Martínez de Aragón et al. (2007) and reported as fungal productivity in dry weight per hectare and year (kg dw ha⁻¹yr⁻¹). Finally, we checked The IUCN Red List of threatened species (IUCN, 2022) to assess the current protection status of the gathered species.

In order to test the effect of the weather conditions on fungal fruiting, we considered the mean minimum, mean and mean maximum monthly temperatures and monthly accumulated rainfall variables from July to November, coinciding with the months in which higher fungal fruiting is detected. In addition, we computed other time-related variables reported in Table 2 and described in Table A1 in Appendix A. All weather variables were created from the daily temperature and accumulated rainfall from 2015 to 2019. These variables were extracted from the weighted mean of interpolated and altitudinally corrected values of different weather variables retrieved from the Meteorological Service of Catalonia (SMC) and the Spanish Meteorological Agency (AEMET) meteorological stations situated within a maximum radius of 50 km surrounding each sampling plot. Namely, daily rainfall and the average, maximum, and minimum daily temperature of each plot were obtained using the R package ‘meteoland’ (De Cáceres et al., 2018).

To analyse the effect of the soil physicochemical properties on the sporocarp community, soil samples were taken in the plots during 2015. After discarding the litter layer, we extracted soil samples (20 × 20 × 20 cm) from the monitoring plots. The soil samples were homogenized and analysed in the laboratory using the methodology described in Alday et al. (2012). Each sample was air-dried and sieved (≤2 mm mesh) to separate the coarse elements fraction from the soil. Sand content was analysed using the Bouyoucos-method (Day, 1965). Soil pH was measured using a conductivity meter in a 1:2.5 soil: deionized water slurry (Allen, 1989). Finally, total organic matter (OM) was analysed using the Walkley-Black method (Walkley, 1947). Only the significant soil variables are described in the text (OM content and pH) and included in Table 1, along with data concerning plot characteristics.

2.3. Data analysis

We used several linear mixed-effects models (LMM) to analyse the effect of the weather conditions on sporocarp productivity and richness and how these two response variables changed over the years. In the LMMs, the sum of sporocarp biomass in dry weight in addition to the sum of sporocarp species number per plot and year was used as a response variable. Following, the weather variables were assigned as fixed effects factors. On the other hand, we considered the plot as

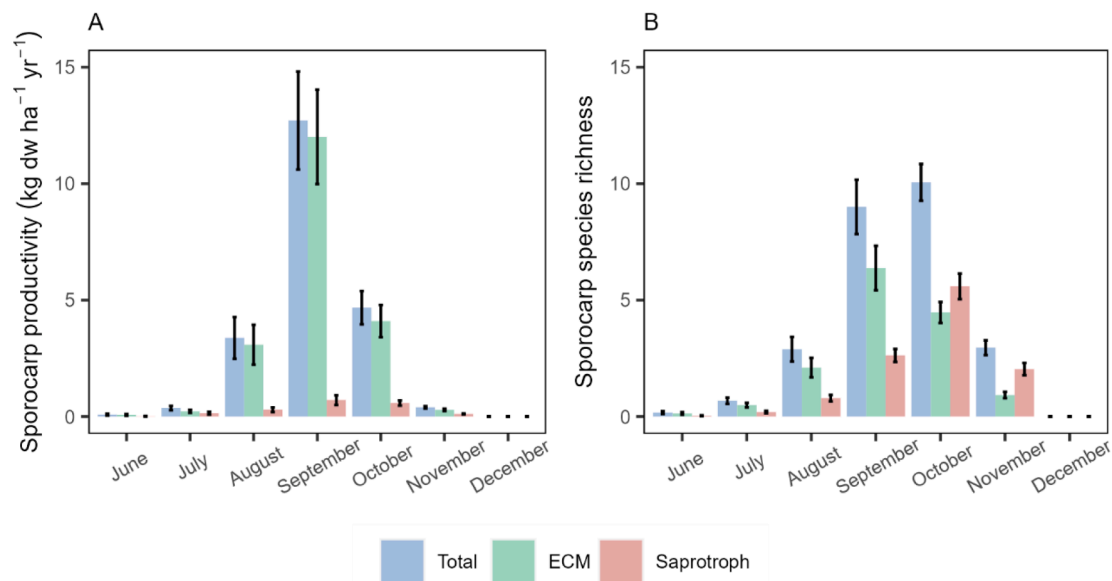


Fig. 2. Mean monthly sporocarp productivity and sporocarp species richness variation along the study period (2015–2019). Sporocarp productivity in dry biomass ($\text{kg dw ha}^{-1} \text{yr}^{-1}$), A. Sporocarp species richness (number of species per plot), B, are split in total (i.e., considering all ectomycorrhizal and saprotrophic macrofungi), ectomycorrhizal, and saprotrophic fungal guilds. Error bars indicate $\pm$ Standard Error (SE) of the mean.

random effects to account for between- and within-plot random variation since the same plots were multiply sampled over time. Evenness was modelled using LMMs, using Pielou's evenness index (Pielou, 1966) as the response variable. Pielou's evenness index was obtained using the dry biomass values of different sporocarp species per plot and year. In the same way that in the sporocarp productivity and richness models, the plot was configured as a random effect on model interception and the weather variables as fixed effects. Sporocarp productivity, species richness and evenness were analysed separately for total, ECM and saprotroph fungal communities. In order to choose the best model, we carefully considered the ecological sense of the variables included in the analysis, secondly, Akaike's information criterion (AIC), and the absence of collinearity (Variance Inflation Factor < 3) between variables using the 'performance' package (Lüdtke et al., 2020).

Thus, sporocarp productivity and richness for total, ECM and saprotrophic fungal communities were obtained using the following formula:

$$(F_c)_{ij} = (\beta + b_{0i}) + \beta X + \varepsilon$$

Where $(F_c)_{ij}$ is the sporocarp productivity or species richness in plot i and year j , β_0 are the fixed effects for each independent variable, b_{0i} denote the random effects on the model intercept, X are vectors of predictor variables while ε is the residual of the model.

Finally, in order to describe the sporocarp community composition across plots and years we run a detrended correspondence analysis (DCA) using the sporocarp matrix of dry biomass. At the same time, i) we tested if such interannual changes were explained by the weather variables and ii) we tested the influence of the soil physicochemical characteristics over the sporocarp community composition between plots. Both analyses were using the passive fit over the previous DCA ordination. All tests were run in three independent response data sets: (i) sporocarps biomass for the total community composition, (ii) sporocarps biomass for the ECM community composition and (iii) sporocarps biomass for the saprotroph community composition to see possible differences between trophic groups in addition of an overall vision of the whole mushroom community composition.

The data analysis was carried out in R software 4.0.2 (R Development Core Team, 2022) using the function 'lmer' implemented in the 'lme4' package (Bates et al., 2015) for fitting the linear mixed-effects models (LMM) and 'vegan' package for multivariate analyses (Oksanen et al.,

2020).

3. Results

3.1. Sporocarp productivity and diversity

The annual productivity of the sampled *Pinus uncinata* stands ranged from 5.45 to 52.72 $\text{kg dw ha}^{-1} \text{yr}^{-1}$ for the total fungal community, with a mean of 21.60 $\text{kg dw ha}^{-1} \text{yr}^{-1}$. For the ECM community, the productivity range was 4.63 to 49.68 $\text{kg dw ha}^{-1} \text{yr}^{-1}$ with a mean of 19.84 $\text{kg dw ha}^{-1} \text{yr}^{-1}$, while for the saprotrophic fungal community, the range was 0.06 to 7.05 $\text{kg dw ha}^{-1} \text{yr}^{-1}$ with a mean of 1.76 $\text{kg dw ha}^{-1} \text{yr}^{-1}$ (Table 2). For the ECM community, the higher annual productivity values were concentrated between August and October, reaching the maximum in September (Fig. 2. A). For the saprotrophic community, we observed a broader range of production months but with lower values, starting in June until November with the minimum productivity in these two months, and smoothly higher productivity values in September (Fig. 2. A).

When examining the sporocarp richness of the studied stands we found a total of 258 species, of which 133 are part of the ECM sporocarp community and 125 are part of the saprotrophic sporocarp community (Table B1 in Appendix B). For the ECM community, the higher sporocarp richness values were reached from August to October, with being September the sporocarp richness peak (Fig. 2. B). Secondly, for the saprotrophic community, the higher sporocarp richness values ranged from September to November, exhibiting higher values during October (Fig. 2. B). On the other hand, we observed varying interannual changes in the evenness of the total, ECM, and saprotrophic fungal communities. The evenness values for the total community ranged from very low to relatively high, with a central tendency around the mean. Similarly, the ECM community exhibited a range of evenness values, indicating fluctuations in its composition over time. Lastly, the evenness of the saprotrophic community displayed a similar pattern, with a range of values indicating varying levels of evenness throughout the study period (Table 2).

When analysing the effect of the weather conditions on the sporocarp fructification we detected a positive and significant effect of the August accumulated rainfall over the total sporocarp richness and evenness. September accumulated rainfall also produced a positive effect on total

Table 3

Statistically significant parameter estimates for the selected models. The response variables tested are sporocarp productivity and sporocarp richness divided into total, ectomycorrhizal (ECM), and saprotrophs. In addition, the sporocarp evenness analysis is displayed for the communities that showed a significant relationship with the environmental variables, namely, total and ECM fungi. Significance levels: p-value < 0.001 ‘***’, p-value < 0.01 ‘**’, p-value < 0.05 ‘*’.

		Model			Sporocarp species richness			Sporocarp species evenness	
		Total	ECM	Saprotrophs	Total	ECM	Saprotroph	Total	ECM
Fixed effects	Intercept	−0.89***	−3.34**	−0.42*	−1.96***	+3.56***	−1.78***	+0.01***	+2.29**
	August rain	–	–	–	+0.01***	–	–	+0.01*	–
	September rain	–	–	–	+0.03***	–	+0.03***	–	–
	October rain	–	–	–	–	–	+0.01***	–	–
	August & September rain	+0.02***	+0.02***	+0.01***	–	+0.02***	–	–	+0.01*
	August min. temp.	–	–	–	–	−0.39***	–	–	−0.19**
	October min. temp.	–	–	–	+0.27***	–	+0.30***	–	–
	November min. temp.	+0.35***	–	–	–	–	–	–	–
	November mean temp.	–	+0.28***	–	–	–	–	–	–
Random effects	SD (intercept)	0.47	0.47	0.26	0.29	0.29	0.36	0.04	0.03
	SD (residual)	0.44	0.52	0.45	–	–	–	0.14	0.23

Abbreviations: rain.: monthly accumulated rainfall (mm); min. temp., mean temp. and max. temp.: mean monthly minimum, mean and maximum temperature (°C), respectively; SD: standard deviation.

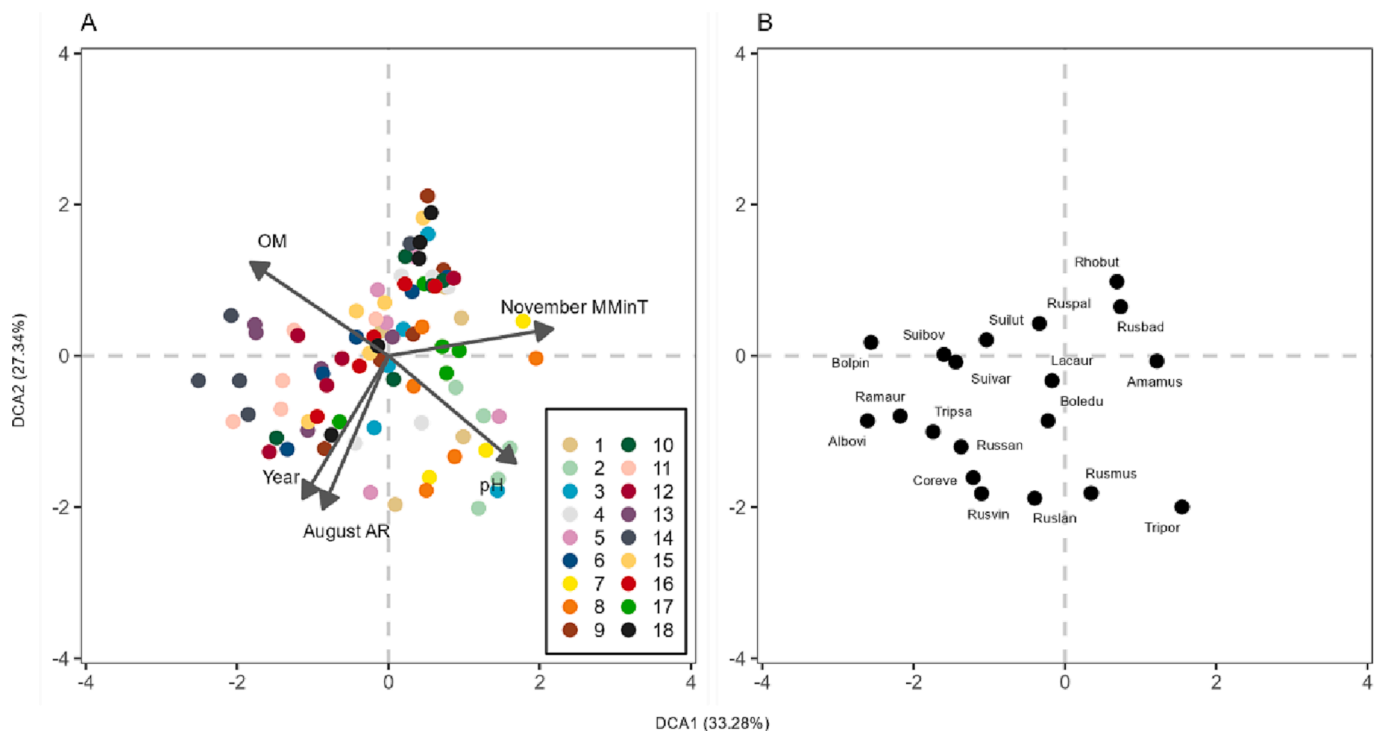


Fig. 3. Detrended correspondence analysis (DCA) of total fungal sporocarp community composition across years in *Pinus uncinata* stands. (A) Sites biplot with different dot colours indicating the belonging plots, and the significant environmental variables overlaid. (B) Species biplot of the most abundant species gathered during the study period. The percentage of variance explained by each DCA-axis is in parentheses. Abbreviations of fungal species shown in the DCA diagrams, Figs. 3-4-5: Albovi – *Albatrellus ovinus*, Amamus – *Amanita muscaria*, Ampcla – *Ampulloclitocybe clavipes*, Boledu – *Boletus edulis*, Bolpin – *Boletus pinophilus*, Clicos – *Clitocybe costata*, Coreve – *Cortinarius evernius*, Entsin – *Entoloma sinuatum*, Gyresc – *Gyromitra esculenta*, Hygaur – *Hygrophoropsis aurantiaca*, Infgib – *Infundibulicybe gibba*, Lacaur – *Lactarius aurantiacus*, Lycper – *Lycoperdon perlatum*, Lyodec – *Lyomices decastes*, Macpro – *Macrolepiota procera*, Melpol – *Melanoleuca polioleuca*, Mycepi – *Mycena epipterygia*, Myclep – *Mycena leptoccephala*, Mycpur – *Mycena pura*, Phogum – *Pholiota gummosa*, Phosqu – *Pholiota squarrosa*, Ramaur – *Ramaria aurea*, Rhobut – *Rhodocollybia butyracea*, Rusbad – *Russula badia*, Ruslan – *Russula langei*, Rusmus – *Russula mustelina*, Ruspai – *Russula paludosa*, Russan – *Russula sanguinea*, Rustur – *Russula turci*, Rusvin – *Russula vinosa*, Scusub – *Scutellinia subhirtella*, Suibov – *Suillus bovinus*, Suilut – *Suillus luteus*, Suivar – *Suillus variegatus*, Tapatr – *Tapinella atrotomentosa*, Tripor – *Tricholoma portentosum*, Tripsa – *Tricholoma psammopus*, Trirut – *Tricholomopsis rutilans*.

and saprotroph sporocarp richness. When grouping both rainfall variables we detected a positive and significant effect on Total, ECM and saprotroph productivity and ECM sporocarp richness and evenness. Moreover, October accumulated rainfall affected positively and significantly saprotroph sporocarp productivity and richness. After analysing the temperature, we detected different effects on the sporocarp fruiting such as August temperatures that affected negatively both ECM

sporocarp richness and evenness. On the other hand, October mean temperatures produced positive and significant effects on the saprotroph richness and similarly, November mean temperatures affected positively the total, ECM and saprotroph sporocarp productivity and total sporocarp richness. However, we did not find any significant relationship between the evenness of the saprotrophic sporocarp community and any of the tested environmental variables (Table 3).

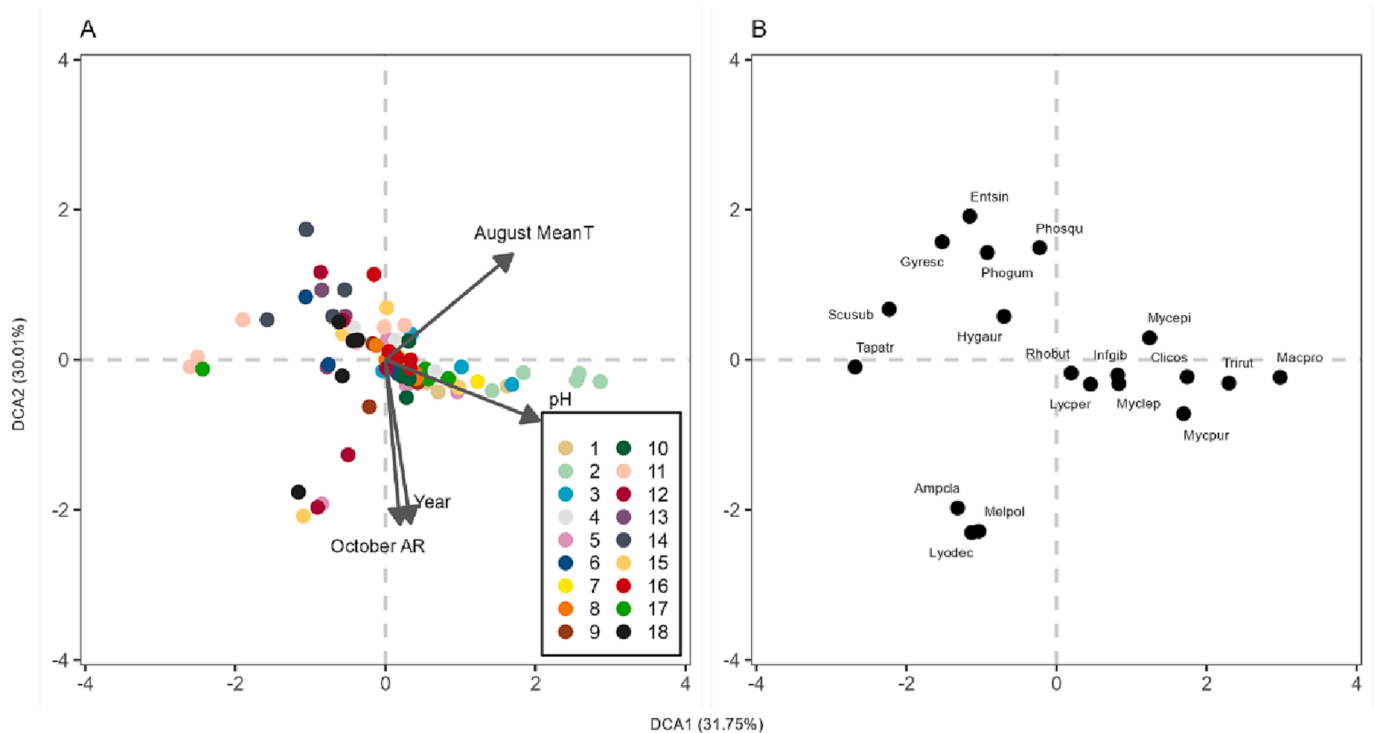


Fig. 4. Detrended correspondence analysis (DCA) of ectomycorrhizal fungal sporocarp community composition across years in *Pinus uncinata* stands. (A) Sites biplot with different dot colours indicating the belonging plots, and the significant environmental variables overlaid. (B) Species biplot of the most abundant species gathered during the study period. The percentage of variance explained by each DCA-axes is in parentheses. Abbreviations of fungal species are shown in the DCA diagrams. See Fig. 3 for species abbreviation.

3.2. Fungal community composition

The ECM sporocarp productivity accounted for 92% of the total productivity with predominant families and corresponding genera such as Russulaceae (44%) with *Russula* (40%) and *Lactarius* (4%), Tricholomataceae (15%) with only *Tricholoma* (15%) and Boletaceae (10%) with genera such as *Suillus* (6%) and *Boletus* (3%). For all the ECM, *Russula* was the most productive genus and included species such as *R. vinosa* (7%), *R. sanguinea* (6%) and *R. lanzei* (2%). The saprotrophic sporocarp productivity contributed 8% of the total productivity with families and corresponding genera such as Omphalotaceae (2%) with *Rhodocollybia* (1.5%) and *Gymnopus* (0.1%), Hygrophoraceae (2%) with *Ampullocitocybe* (0.9%) and *Hygrophorus* (0.5%), and finally Lyophyllaceae (1%) with *Lyophyllum* (0.8%). *Rhodocollybia* was the most productive genus and was represented by *R. butyracea* (1.5%) and one observation of *R. prolixa* (0.01%).

The DCA ordination for the total community composition produced eigenvalues (λ) 0.5730, 0.4706, 0.3784 and 0.2995, and GL of 4.4624, 4.1278, 3.5616 and 3.3777 for the first four axes respectively. The ordination showed that sporocarp composition varied slightly across plots although most of the plots are in the central area of the ordination (Fig. 3. A). November mean minimum temperature variable (November MMinT; explained variation = 20%, $P = 0.027$) separated plot 1, 2, 7 and 8 with higher values in the positive end of the DCA1 and plot 11 in the negative end with lower temperatures. The rest of the plots did not follow a distribution among the axes. DCA2 was separated on the positive-end by OM content (explained variation = 16%, $P = 0.049$), while the negative-end was defined by August accumulated rainfall (August AR) (explained variation = 60%, $P = 0.001$) and pH (explained variation = 71%, $P = 0.001$). The studied plots showed no clear trend or preference for the studied explicative variables. There is a significant effect of the Year variable (explained variation = 41%, $P = 0.001$) over the total sporocarp community composition although we did not observe clear trends on how species are distributed during the study

period.

Relative abundances of certain species such as *Amanita muscaria*, *Rhodocollybia butyracea* and *Russula badia* were related to higher November MMinT (Fig. 3. B), and other species such as *Albatrellus ovinus*, *Ramaria aurea* or *Tricholomopsis rutilans* were more related to lower temperatures. Regarding August accumulated rainfall (August AR), *Cortinarius evernius* and some *Russula* species such as *R. lanzei* and *R. vinosa* preferred higher rainfall during this month. Higher values of pH were related to species such as *Tricholoma portentosum* while *Boletus edulis* preferred lower pH values. Finally, *Suillus* species such as *S. luteus*, *S. bovinus* and *S. variegatus* preferred plots with a certain amount of OM.

The DCA ordination for ECM community composition produced eigenvalues (λ) of 0.5705, 0.4605, 0.2643 and 0.2991, and GL of 4.3641 4.1457, 4.3517 and 3.6248 for the first four axes respectively. ECM sporocarp ordination showed similar results as the total community DCA ordination since the relative abundance of the species was related to November MMinT (explained variation = 22%, $P = 0.024$), pH (explained variation = 67%, $P = 0.001$) and August accumulated rainfall (August AR) (explained variation = 40%, $P = 0.001$) (Fig. 4. A). Relative abundances of species such as *Russula badia*, *R. paludosa* and *Amanita muscaria* are related to higher November MMinT while *Albatrellus ovinus* or *Ramaria aurea* are related to lower November MMinT. Other species such as *Tricholoma portentosum* were related to higher pH values while on the opposite side, species belonging to the *Suillus* genus appeared to be related to lower pH values. Moreover, some *Russula* species such as *R. vinosa* or *R. sanguinea* and *Tricholoma psammopus* were related to higher August accumulated rainfall (August AR) values (Fig. 4. B) while *R. badia* and *Amanita muscaria* were related to lower values of this variable. In the same way as total sporocarp community composition, we detected a significant year effect (explained variation = 27%, $P = 0.009$) in the community analysis although we could not identify a species community trend during the study period (Fig. 4. A).

The saprotrophic sporocarps DCA (eigenvalues (λ) of 0.8161, 0.77132, 0.4665 and 0.5162, and GL of 5.4644, 3.8239, 3.3934, and

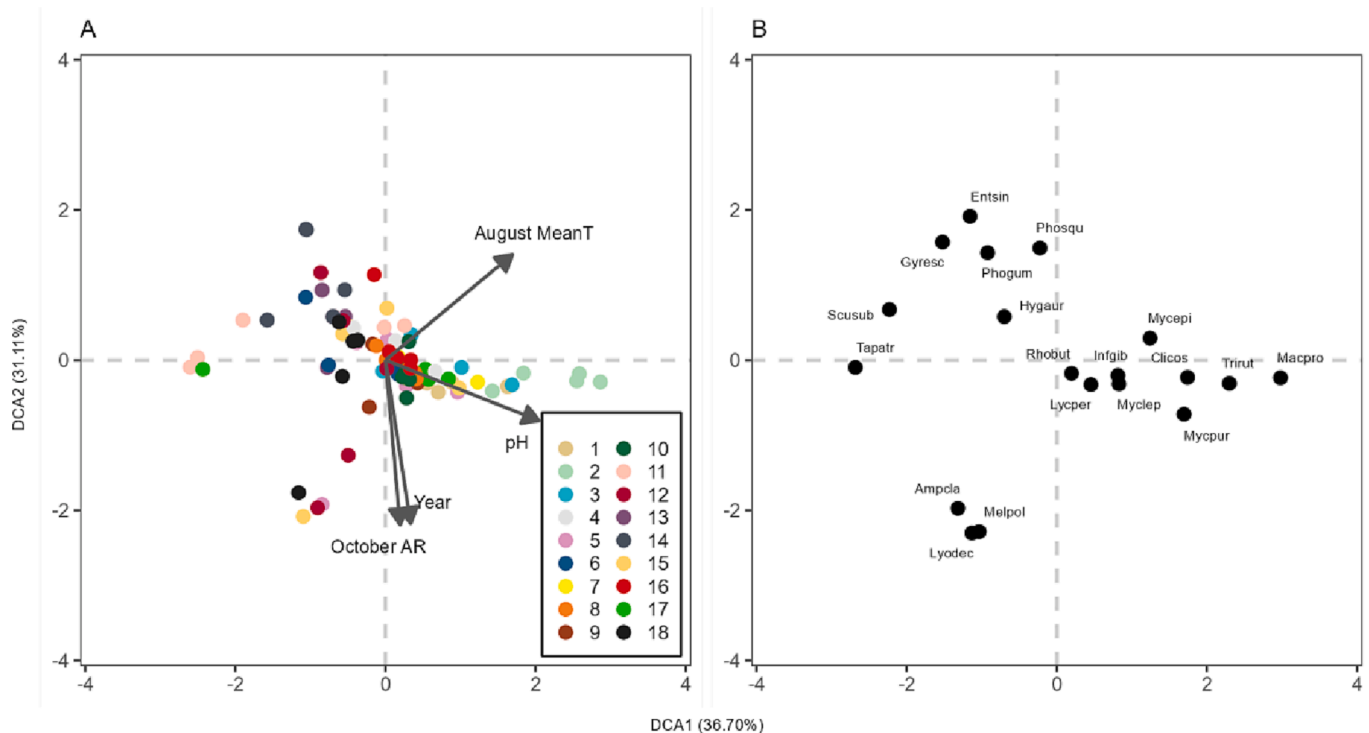


Fig. 5. Detrended correspondence analysis (DCA) of saprotrophic fungal sporocarp community composition across years in *Pinus uncinata* stands. (A) Sites biplot with different dot colours indicating the belonging plots, and the significant environmental variables overlaid. (B) Species biplot of the most abundant species gathered during the study period. The percentage of variance explained by each DCA-axis is in parentheses. Abbreviations of fungal species are shown in the DCA diagrams. See Fig. 3 for species abbreviation.

Table 4

List of the species gathered from 2015 to 2019 included in the IUCN.

	Year					Category
	2015	2016	2017	2018	2019	
<i>Agaricus sylvaticus</i>	–	–	–	1	–	LC
<i>Boletus edulis</i>	7	1	–	18	1	LC
<i>Boletus pinophilus</i>	2	2	2	7	–	LC
<i>Clitopilus prunulus</i>	10	2	–	12	–	LC
<i>Cortinarius caperatus</i>	2	–	–	9	–	LC
<i>Gomphidius roseus</i>	–	–	–	1	–	LC
<i>Hygrophorus hypothejus</i>	29	13	2	6	2	LC
<i>Imleria badia</i>	–	–	–	6	–	LC
<i>Lycoperdon perlatum</i>	15	6	3	20	1	LC
<i>Russula aeruginea</i>	1	–	–	–	–	LC
<i>Russula paludosa</i>	6	2	1	8	–	LC
<i>Russula vinosa</i>	1	–	–	66	–	LC
<i>Suillus bovinus</i>	13	8	2	19	1	LC
<i>Suillus granulatus</i>	–	–	–	–	–	LC
<i>Suillus granulatus</i>	2	2	–	1	1	LC
<i>Suillus luteus</i>	10	8	–	18	–	LC
<i>Suillus variegatus</i>	29	13	3	4	–	LC
<i>Entoloma porphyrophaeum</i>	2	–	3	1	–	VU

Abbreviations: LC: Least Concern; VU: Vulnerable.

3.5022) showed that the passive overlaid of August mean temperature (August MeanT) (explained variation = 37%, $P = 0.009$), pH (explained variation = 54%, $P = 0.001$) and October accumulated rainfall (October AR) (explained variation 31%, $P = 0.023$) were significant (Fig. 5. A). The relative abundance of *Rhodocollybia butyracea* was related to lower August mean temperatures. Higher values of pH were related to species such as *Mycena pura* or *M. leptcephala* while on the negative part of the DCA2 axis we found species such as *Tapinella atrotomentosa*, *Gyromitra esculenta* or *Scutellinia subhirtella* which were related to lower pH values. When checking October AR, species such as *Cystoderma amianthinum* were related to higher accumulated rainfall values and other species

such as *Entoloma sinapizans* or *Pholiota gummosa* were related to lower October AR values (Fig. 5. B). In addition, the significant effect of the Year variable is lacking in the saprotroph sporocarp community (explained variation = 18%, $P = 0.143$).

Finally, comparing our data set with the IUCN Red List, we found 17 species growing in our *Pinus uncinata* stands and included in the categories Least Concern (LC), with 16 species and Vulnerable (VU), with 1 species (Table 4). The majority of species listed as LC fructified in different years without a clear increasing or decreasing trend. Other species accounted for just one observation in the whole study period, such as *Gomphidius roseus*, while others, such as *Imleria badia*, appeared multiple times but only during one single year.

4. Discussion

4.1. Sporocarp productivity and diversity

This study reveals a high interannual variation in mushroom fruiting, remarking the high mean productivity values of the studied stands compared with other *Pinus* ecosystems in the Iberian Peninsula (Karavani et al., 2018; Alday et al., 2017; de-Miguel et al., 2014; Martínez-Peña et al., 2012). A possible explanation for the detected productivity values is the high rainfall regime in our study area, which is mostly concentrated during late summer and until October, compared to other *Pinus* species stands located at lower latitudes with lower rainfall regimes (de-Miguel et al., 2014). Another feature explaining the mushroom productivity in our study area is temperature, as previous studies have reported that cold weather and frosts in high-elevation and continental conditions can reduce productivity (de-Miguel et al., 2014; Martínez-Peña et al., 2012). In our studied stands, lower temperatures typical of high altitudes promote both an early start of sporocarp fruiting during late summer and an early end of fungal fruiting during October–November (in both cases, earlier than in other forest ecosystems at lower altitudes).

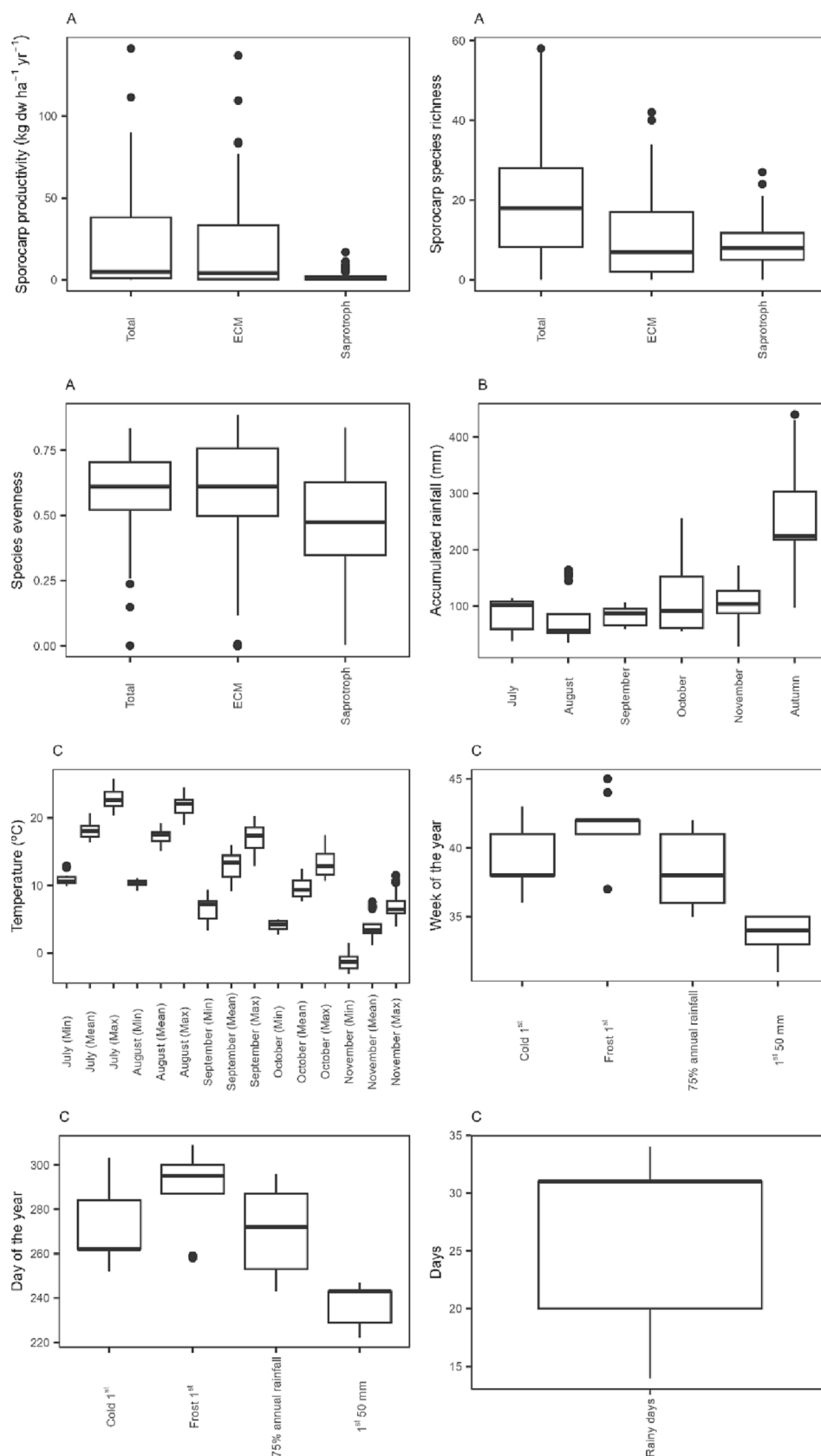


Fig. A1. Plots of the modelling data.

Table A1

Description of the modelling data per plot and year from 2015 to 2019.

Total productivity (dry biomass in kg dw ha ⁻¹)	Accumulated total community dry weight
ECM productivity (dry biomass in kg dw ha ⁻¹)	Accumulated ECM community dry weight
Saprotrophic productivity (dry biomass in kg dw ha ⁻¹)	Accumulated saprotrophic community dry weight
Total species richness	Total community species number
ECM species richness	Ectomycorrhizal community species number
Saprotrophic species richness	Saprotrophic community species number
July rainfall (mm)	July accumulated rainfall
August rainfall (mm)	August accumulated rainfall
September rainfall (mm)	September accumulated rainfall
October rainfall (mm)	October accumulated rainfall
November rainfall (mm)	November accumulated rainfall
1st 50 mm (Week / DOY)	The date from the 1st of August, in which 50 mm of rainfall is reached
75% annual rainfall (Week / DOY)	The date on which 75% of the annual accumulated rainfall is reached
Autumn rainfall (mm)	Accumulated rainfall between the 1st of August and the first day on which temperatures fell below 0 °C.
Number of rainy days	Number of rainy days from the 1st of August until the first day on which temperatures fell below 0 °C.
July min. temperature (°C)	Mean minimum July temperature
July mean temperature (°C)	Mean July temperature
July max. temperature (°C)	Mean maximum July temperature
August min. temperature (°C)	Mean minimum August temperature
August mean temperature (°C)	Mean August temperature
August max. temperature (°C)	Mean maximum August temperature
September min. temperature (°C)	Mean minimum September temperature
September mean temperature (°C)	Mean September temperature
September max. temperature (°C)	Mean maximum September temperature
October min. temperature (°C)	Mean minimum October temperature
October mean temperature (°C)	Mean October temperature
October max. temperature (°C)	Mean maximum October temperature
November min. temperature (°C)	Mean minimum November temperature
November mean temperature (°C)	Mean November temperature
November max. temperature (°C)	Mean maximum November temperature
Cold 1st (Week / DOY)	The date, from the 1st of August in Julian date, in which temperatures below 5 °C were registered during 5 consecutive days
Frost 1st (Week / DOY)	The date, from the 1st of August in Julian date, in which temperatures below 5 °C were registered during 5 consecutive days

Abbreviations. ECM: ectomycorrhizal; Week: week of the year; DOY: day of the year.

In analysing productivity outcomes across trophic groups, we observed variation in both the timing and quantity of biomass generated. Specifically, the ECM group exhibited a pronounced production pattern in ECM fruiting, with significant differences between months. Mushroom emergence in this group started in late summer, reached its peak in September, and declined sharply until November. Conversely, saprotroph fruiting manifested a relatively even production pattern, with negligible differences between months resulting in nearly consistent biomass production throughout the fruiting season with stable productivity values. These results suggest a more compressed annual fruiting period for ECM fungi since they rely on carbohydrate supply from their hosts as reported by [Kausar et al. \(2012\)](#) in a European study concerning data from Austria, Norway, Switzerland and the UK, while saprotrophs may be more directly influenced by weather conditions and, consequently, more dependent on the environment. Furthermore, mixed-effect models showed that total and ECM sporocarp

Table B1

List of the ectomycorrhizal (ECM) and saprotroph (S) species found during the study period.

Species	Group	Abundance (%)
<i>Agaricus sylvaticus</i> Schaeff.	S	0.01
<i>Agrocybe pediades</i> (Fr.) Fayod	S	0.00
<i>Albatrellus ovinus</i> (Schaeff.) Kotl. & Pouzar	ECM	8.90
<i>Amanita battarrae</i> (Boud.) Bon	ECM	0.15
<i>Amanita citrina</i> (Sch.) Pers.	ECM	0.06
<i>Amanita eliae</i> Quél.	ECM	0.04
<i>Amanita excelsa</i> (Fr.) Bertill.	ECM	0.29
<i>Amanita fulva</i> Fr.	ECM	0.27
<i>Amanita gemmata</i> (Fries) Bert.	ECM	0.07
<i>Amanita gracilior</i> Bas & Honrubia	ECM	0.01
<i>Amanita muscaria</i> (L.) Hook.	ECM	1.25
<i>Amanita pantherina</i> (DC.) Krombh	ECM	0.64
<i>Amanita rubescens</i> Pers.	ECM	0.02
<i>Amanita</i> sp.	ECM	0.02
<i>Amanita vaginata</i> (Bull.) Vitt.	ECM	0.02
<i>Ampulloclitocybe clavipes</i> (Pers.) Redhead, Lutzoni, Moncalvo & Vilgalys	S	0.93
<i>Arrhenia obscurata</i> (D.A. Reid) Redhead, Lutzoni, Moncalvo & Vilgalys	S	0.00
<i>Arrhenia rickenii</i> (Hora) Watling	S	0.00
<i>Atheniella flavoalba</i> (Fr.) Redhead, Moncalvo, Vilgalys, Desjardin & B.A. Perry	S	0.00
<i>Atractosporocybe inornata</i> (Sowerby) P. Alvarado, G. Moreno & Vizzini	S	0.00
<i>Auriscalpium vulgare</i> Gray	S	0.00
<i>Baeospora myosura</i> (Fr.) Singer	S	0.01
<i>Boletus edulis</i> Bull.	ECM	2.22
<i>Boletus pinophilus</i> Pilát & Dermek	ECM	1.22
<i>Bovista plumbea</i> Pers.	S	0.00
<i>Calocybe gangraenosa</i> (Fr.) V. Hofst., Moncalvo, Redhead & Vilgalys	S	0.02
<i>Calvatia</i> sp.	S	0.00
<i>Cantharellus cibarius</i> Fr.	ECM	0.39
<i>Chroogomphus helveticus</i> (Singer) M.M. Moser	ECM	0.44
<i>Chroogomphus rutilus</i> (Schaeff.) O.K. Mill.	ECM	0.26
<i>Clavariadelphus pistillaris</i> (L.) Donk	S	0.00
<i>Clitocybe agrestis</i> Harmaja	S	0.00
<i>Clitocybe brumalis</i> (Fr.) Quél.	S	0.00
<i>Clitocybe costata</i> Kühner & Romagn.	S	0.32
<i>Clitocybe dicolor</i> (Pers.) Murrill	S	0.01
<i>Clitocybe ditopa</i> (Fr.) Gillet	S	0.01
<i>Clitocybe fragrans</i> (With.) P. Kumm.	S	0.00
<i>Clitocybe hydrogramma</i> (Bull.) P. Kumm.	S	0.00
<i>Clitocybe metachroa</i> (Fr.) P. Kumm.	S	0.01
<i>Clitocybe nebularis</i> (Batsch. ex Fr.) Kumm.	S	0.02
<i>Clitocybe paulensis</i> Henn.	S	0.00
<i>Clitocybe perforata</i> Corner	S	0.00
<i>Clitocybe phaeophthalma</i> (Pers.) Kuyper	S	0.01
<i>Clitocybe phyllophila</i> (Pers.) P. Kumm.	S	0.03
<i>Clitocybe rivulosa</i> (Pers.) P. Kumm.	S	0.01
<i>Clitocybe robusta</i> Peck.	S	0.00
<i>Clitocybe</i> sp.	S	0.20
<i>Clitocybe vibecina</i> (Fr.) Quél.	S	0.02
<i>Clitocybe dealbata</i> (Sowerby) P. Kumm.	S	0.00
<i>Clitopilus prunulus</i> (Scop.) P. Kumm.	ECM	0.06
<i>Conocybe pulchella</i> (Velen.) Hauskn. & Svrček	S	0.00
<i>Conocybe tenera</i> (Schaeff.) Kühner	S	0.01
<i>Contumyces rosellus</i> (M.M. Moser) Redhead, Moncalvo, Vilgalys & Lutzoni	S	0.00
<i>Cortinarius acutus</i> (Pers.) Fr.	ECM	0.00
<i>Cortinarius anomalus</i> (Fr.) Fr.	ECM	0.05
<i>Cortinarius caeruleus</i> (Sch.) Fr.	ECM	0.04
<i>Cortinarius caninus</i> (Fr.) Fr.	ECM	0.26
<i>Cortinarius caperatus</i> (Pers.) Fr.	ECM	0.47
<i>Cortinarius cinnamomeolutes</i> P.D. Orton	ECM	0.01
<i>Cortinarius cinnamomeus</i> (L.) Gray	ECM	0.39
<i>Cortinarius decipiens</i> (Pers.) Zawadzki	ECM	0.34
<i>Cortinarius elegantior</i> (Fr.) Fr.	ECM	0.42
<i>Cortinarius eustratulus</i> Rob. Henry	ECM	0.01
<i>Cortinarius evernius</i> (Fr.) Fr.	ECM	1.27
<i>Cortinarius flexipes</i> (Pers.) Fr.	ECM	0.02
<i>Cortinarius fulgens</i> Fr.	ECM	0.13

(continued on next page)

Table B1 (continued)

Species	Group	Abundance (%)
<i>Cortinarius fulvescens</i> Fr.	ECM	0.01
<i>Cortinarius infractus</i> (Pers.) Fr.	ECM	0.06
<i>Cortinarius laccatus</i> Reumaux	ECM	0.01
<i>Cortinarius malachius</i> (Fr.) Fr.	ECM	0.00
<i>Cortinarius mucosus</i> (Bull.) J. Kickx f.	ECM	0.73
<i>Cortinarius obtusus</i> Fr.	ECM	0.09
<i>Cortinarius olearioides</i> Rob. Henry	ECM	0.36
<i>Cortinarius pratensis</i> (Bon & Gaugué) Høil.	ECM	0.01
<i>Cortinarius privignus</i> (Fr.) Fr.	ECM	0.32
<i>Cortinarius pulchripes</i> Favre.	ECM	0.01
<i>Cortinarius sanguineus</i> (Wulfen) Gray	ECM	0.06
<i>Cortinarius sphagnogenus</i> (M.M. Moser) Nezdobjm.	ECM	0.00
<i>Cortinarius</i> sp.	ECM	2.53
<i>Cortinarius traganus</i> (Fr.) Fr.	ECM	0.03
<i>Cortinarius trivialis</i> J.E. Lange	ECM	0.00
<i>Crinipellis scabella</i> (Alb. & Schwein.) Murrill	S	0.00
<i>Cystoderma amianthinum</i> (Scop.) Fayod	S	0.06
<i>Cystoderma carcharias</i> (Pers.) Fayod	S	0.02
<i>Cystodermella terryi</i> (Berk. & Broome) Bellù	S	0.03
<i>Deconica coprophila</i> (Bull.) P. Karst.	S	0.00
<i>Deconica merdaria</i> (Fr.) Noordel.	S	0.00
<i>Deconica montana</i> (Pers.) P.D. Orton	S	0.00
<i>Entocybe nitida</i> (Quél.) T.J. Baroni, Largent & V. Hofst.	S	0.00
<i>Entoloma araneosum</i> (Quél.) M.M. Moser	S	0.01
<i>Entoloma cetratum</i> (Fr.) M.M. Moser	S	0.01
<i>Entoloma conferendum</i> (Britzelm.) Noordel.	S	0.01
<i>Entoloma elodes</i> (Fr.) P. Kumm.	S	0.00
<i>Entoloma papillatum</i> (Bres.) Dennis	S	0.00
<i>Entoloma porphyrophaeum</i> (Fr.) P. Karst.	S	0.01
<i>Entoloma rhodopolium</i> (Fr.) P. Kumm.	S	0.01
<i>Entoloma sericeum</i> Quél.	S	0.00
<i>Entoloma sinuatum</i> (Bull.) P. Kumm.	S	0.63
<i>Entoloma</i> sp.	S	0.01
<i>Galerina praticola</i> (F.H. Møller) P.D. Orton	S	0.00
<i>Galerina sideroides</i> (Bull.) Kühner	S	0.01
<i>Galerina</i> sp.	S	0.00
<i>Galerina sphagnorum</i> (Pers.) Kühner	S	0.00
<i>Galerina stylifera</i> (G.F. Atk.) A.H. Sm. & Singer	S	0.00
<i>Gomphidius roseus</i> (Fr.) Oudem.	ECM	0.00
<i>Gymnopilus penetrans</i> (Fr.) Murrill	S	0.02
<i>Gymnopilus stabilis</i> (Weinm.) Kühner & Romagn. ex Bon	S	0.00
<i>Gymnopus androsaceus</i> (L.) Della Magg. & Trassin.	S	0.03
<i>Gymnopus dryophilus</i> (Bull.) Murrill	S	0.00
<i>Gymnopus erythropus</i> (Pers.) Antonín, Halling & Noordel.	S	0.02
<i>Gyromitra esculenta</i> Pers. ex Fr.	S	0.13
<i>Hebeloma crustuliniforme</i> (Bull.) Quél.	ECM	0.02
<i>Hebeloma laterinum</i> (Batsch) Vesterh.	ECM	0.02
<i>Hebeloma mesophaeum</i> (Pers.) Quél.	ECM	0.00
<i>Hebeloma sinapizans</i> (Paulet) Gillet	ECM	0.04
<i>Hebeloma</i> sp.	ECM	0.01
<i>Hemimycena mairei</i> (Gilb.) Sing.	S	0.00
<i>Hydnellum aurantiacum</i> (Batsch: Fr.) P. Karst.	ECM	0.02
<i>Hydnellum scabrosum</i> (Fr.) E. Larss., K.H. Larss. & Köljalg	ECM	0.03
<i>Hygrophoropsis aurantiaca</i> (Wulfen) Maire	S	0.61
<i>Hygrophorus hypothecus</i> (Fr.) Fr.	ECM	0.50
<i>Hypholoma elongatum</i> (Pers.) Ricken	S	0.00
<i>Hypholoma fasciculare</i> (Huds.) Kumm.	S	0.00
<i>Hypholoma lateritium</i> (Schaeff.) P. Kumm	S	0.00
<i>Hypholoma marginatum</i> J. Schröt.	S	0.00
<i>Imleria badia</i> (Fr.) Vizzini	ECM	0.33
<i>Infundibulicybe gibba</i> (Pers.) Harmaja	S	0.17
<i>Inocybe fuscicula</i> Velen	ECM	0.00
<i>Inocybe glabripes</i> Ricken	ECM	0.03
<i>Inocybe lacera</i> (Fr.) P. Kumm.	ECM	0.00
<i>Inocybe phaeodisca</i> Kühn.	ECM	0.01
<i>Inocybe praetervisa</i> Quél.	ECM	0.00
<i>Inocybe pyriodora</i> (Pers.) P. Kumm.	ECM	0.00
<i>Inocybe</i> sp.	ECM	0.00
<i>Inosperma erubescens</i> (A. Blytt) Matheny & Esteve-Rav.	ECM	0.01
<i>Laccaria laccata</i> (Scop.) Cooke	ECM	0.02
<i>Lacrymaria lacrymabunda</i> (Bull.) Pat.	S	0.00
<i>Lactarius atlanticus</i> Bon	ECM	0.10
<i>Lactarius aurantiacus</i> (Pers.) Gray	ECM	2.11
<i>Lactarius chrysorrheus</i> Fr.	ECM	0.01
<i>Lactarius deliciosus</i> (L.) Gray	ECM	0.12

Table B1 (continued)

Species	Group	Abundance (%)
<i>Lactarius deterrimus</i> Gröger	ECM	0.79
<i>Lactarius glyciosmus</i> (Fr.) Fr.	ECM	0.00
<i>Lactarius hepaticus</i> Plowr.	ECM	0.01
<i>Lactarius lignyotus</i> Fr.	ECM	0.02
<i>Lactarius picinus</i> Fr.	ECM	0.08
<i>Lactarius rufus</i> (Scop.) Fr.	ECM	0.53
<i>Lactarius semisanguifluus</i> R. Heim & Leclair	ECM	0.01
<i>Lactarius trivialis</i> (Fr.) Fr.	ECM	0.00
<i>Lactifluus volemus</i> (Fr.) Kuntze	ECM	0.02
<i>Lepiota castanea</i> Quél.	S	0.00
<i>Lepiota magnispora</i> Murrill	S	0.05
<i>Lepiota</i> sp.	S	0.06
<i>Leratiomyces squamosus</i> (Pers.) Bridge & Spooner	S	0.01
<i>Lichenomphalia umbellifera</i> (L.) Redhead, Lutzoni, Moncalvo & Vilgalys	S	0.01
<i>Lycoperdon echinatum</i> Pers.	S	0.00
<i>Lycoperdon perlatum</i> Pers.	S	0.17
<i>Lycoperdon</i> sp.	S	0.00
<i>Lyophyllum decastes</i> (Fr.) Sing.	S	0.77
<i>Lyophyllum</i> sp.	S	0.00
<i>Macrolepiota procera</i> (Scop.) Sing.	S	0.43
<i>Marasmius rotula</i> (Scop.) Fr.	S	0.00
<i>Marasmius scopulatus</i> Desjardin & Ovrebø	S	0.00
<i>Marasmius</i> sp.	S	0.00
<i>Marasmius wynneae</i> Berk. & Broome	S	0.00
<i>Melanoleuca polioleuca</i> (Fr.) Kühner & Maire	S	0.07
<i>Melanoleuca pseudoluscina</i> Bon	S	0.00
<i>Mycena acicula</i> (Schaeff.) P. Kumm.	S	0.00
<i>Mycena aetites</i> (Fr.) Quél.	S	0.00
<i>Mycena alcalina</i> (Fr.) Kumm.	S	0.03
<i>Mycena arcangeliana</i> Bres.	S	0.00
<i>Mycena aurantiaca</i> (Murrill) Murrill	S	0.01
<i>Mycena capillaripes</i> Peck	S	0.00
<i>Mycena crocata</i> (Schrad.) P. Kumm.	S	0.00
<i>Mycena epipterygia</i> (Scop.) Gray	S	0.06
<i>Mycena filipes</i> (Bull.) Kumm.	S	0.00
<i>Mycena flavescens</i> Vel.	S	0.00
<i>Mycena galericulata</i> (Scop.) Gray	S	0.02
<i>Mycena leptcephala</i> (Pers.) Gill.	S	0.08
<i>Mycena metata</i> (Fr.) P. Kumm.	S	0.00
<i>Mycena pelianthina</i> (Fr.) Quél.	S	0.00
<i>Mycena polygramma</i> (Bull.) Gray	S	0.00
<i>Mycena pura</i> (Pers.) P. Kumm.	S	0.10
<i>Mycena rosella</i> (Fr.) P. Kumm.	S	0.00
<i>Mycena sanguinolenta</i> (Alb. & Schwein.) P. Kumm.	S	0.00
<i>Mycena seymii</i> Quél.	S	0.00
<i>Mycena</i> sp.	S	0.04
<i>Mycena vitilis</i> (Fr.) Quél.	S	0.00
<i>Mycena vulgaris</i> (Pers.) P. Kumm.	S	0.00
<i>Mycena zephirus</i> (Fr.) P. Kumm.	S	0.00
<i>Mycenella bryophila</i> (Voglino) Singer	S	0.02
<i>Mycetinus scorodoni</i> (Fr.) A.W. Wilson & Desjardin	S	0.01
<i>Myochromella boudieri</i> (Kühner & Romagn.) V. Hofst., Clémenceçon, Moncalvo & Redhead	S	0.00
<i>Neoboletus erythropus</i> (Pers.) C. Hahn	ECM	0.14
<i>Otidea alutacea</i> (Pers.) Massee	ECM	0.08
<i>Panaeolus semiovatus</i> (Sowerby) S. Lundell & Nannf.	S	0.00
<i>Paragymnopus perforans</i> (Hoffm.) J.S. Oliveira	S	0.01
<i>Phaeocollybia lugubris</i> (Fr.) R. Heim	S	0.04
<i>Phellodon tomentosus</i> (L.) Banker	ECM	0.04
<i>Phlegmacium glaucopus</i> (Schaeff.) Wünsche	ECM	0.11
<i>Pholiota gummosa</i> (Lasch) Singer	S	0.07
<i>Pholiota squarrosa</i> (Vahl) P. Kumm.	S	0.10
<i>Pluteus cinereofuscus</i> J.E. Lange	S	0.00
<i>Pluteus phlebophorus</i> (Ditmar) P. Kumm.	S	0.01
<i>Polyporus</i> sp.	S	0.01
<i>Psathyrella spadiceogrisea</i> (Schaeff.) Maire	S	0.00
<i>Pseudoclitocybe expallens</i> (Pers.) M.M. Moser	S	0.01
<i>Ramaria aurea</i> (Sch.) Quél.	ECM	1.09
<i>Ramaria stricta</i> (Pers.) Quél.	S	0.00
<i>Rhizopogon luteolus</i> Fr.	ECM	0.04
<i>Rhizopogon</i> sp.	ECM	0.00
<i>Rhodocollybia butyracea</i> (Bull.) Lennox	S	1.51
<i>Rhodocollybia proluxa</i> (Fr.) Antonín & Noordel.	S	0.00
<i>Russula adusta</i> (Pers.) Fr.	ECM	0.04

(continued on next page)

Table B1 (continued)

Species	Group	Abundance (%)
<i>Russula aeruginea</i> Lindblad ex Fr.	ECM	0.07
<i>Russula albonigra</i> (Krombh.) Fr.	ECM	0.02
<i>Russula badia</i> Quél.	ECM	1.64
<i>Russula caerulea</i> Fr.	ECM	0.01
<i>Russula cessans</i> A. Pearson	ECM	0.06
<i>Russula chloroides</i> (Krombh.) Bres.	ECM	0.92
<i>Russula decipiens</i> (Singer) Bon	ECM	0.02
<i>Russula delicata</i> Fr.	ECM	0.50
<i>Russula emetica</i> (Schaeff.) Pers.	ECM	0.12
<i>Russula exalbicans</i> (Pers.) Melzer & Zvára	ECM	0.10
<i>Russula foetens</i> Pers.	ECM	0.13
<i>Russula fragilis</i> Fr.	ECM	0.03
<i>Russula heterophylla</i> (Fr.) Fr.	ECM	0.20
<i>Russula integra</i> (L.) Fr.	ECM	0.01
<i>Russula langei</i> Bon	ECM	2.34
<i>Russula mustelina</i> Fr.	ECM	2.11
<i>Russula paludosa</i> Britzelm.	ECM	1.77
<i>Russula parazurea</i> Jul. Schäff.	ECM	0.01
<i>Russula queletii</i> Fr.	ECM	0.44
<i>Russula romellii</i> Maire	ECM	0.67
<i>Russula sanguinea</i> Fr.	ECM	6.19
<i>Russula sardonica</i> Fr.	ECM	0.13
<i>Russula</i> sp.	ECM	14.60
<i>Russula torulosa</i> Bres.	ECM	0.14
<i>Russula turci</i> Bres.	ECM	0.97
<i>Russula vinosa</i> Lindblad	ECM	6.90
<i>Russula virescens</i> (Sch.) Fr.	ECM	0.01
<i>Russula xerampelina</i> (Schaeff.) Fr.	ECM	0.15
<i>Scleroderma areolatum</i> Ehrenb.	ECM	0.02
<i>Scutellinia subhirtella</i> Svrček	S	0.09
<i>Suillus bovinus</i> (L.) Roussel	ECM	1.34
<i>Suillus collinitus</i> (Fr.) Kuntze	ECM	0.00
<i>Suillus granulatus</i> (L.) Roussel	ECM	0.12
<i>Suillus luteus</i> (L.) Roussel.	ECM	1.10
<i>Suillus</i> sp.	ECM	0.07
<i>Suillus variegatus</i> (Sw.) Richon & Roze	ECM	3.46
<i>Tapinella atrotomentosa</i> (Batsch) Šutara	S	0.52
<i>Tapinella panuoides</i> (Fr.) E.-J. Gilbert	S	0.01
<i>Tephroclype rancida</i> (Fr.) Donk	S	0.00
<i>Thaxterogaster causticus</i> (Fr.) Niskanen & Liimat.	ECM	0.01
<i>Thaxterogaster purpurascens</i> (Fr.) Niskanen & Liimat.	ECM	0.62
<i>Thelephora terrestris</i> Ehrh.	ECM	0.01
<i>Tricholoma argenteum</i> Ovrebo	ECM	0.00
<i>Tricholoma aurantium</i> (Schaeff.) Ricken	ECM	0.17
<i>Tricholoma equestre</i> (L.) P. Kumm.	ECM	0.49
<i>Tricholoma fracticum</i> (Britzelm.) Kreisel	ECM	0.70
<i>Tricholoma imbricatum</i> (Fr.) Kumm.	ECM	0.09
<i>Tricholoma josserandii</i> Bon	ECM	0.15
<i>Tricholoma pessundatum</i> (Fr.) Quél.	ECM	0.01
<i>Tricholoma portentosum</i> (Fr.) Quél.	ECM	11.48
<i>Tricholoma psammopus</i> (Kalchbr.) Quél.	ECM	0.98
<i>Tricholoma sejunctum</i> (Sowerby) Quél.	ECM	0.14
<i>Tricholoma</i> sp.	ECM	0.02
<i>Tricholoma sparsifolium</i> Velen.	ECM	0.02
<i>Tricholoma sulphurellum</i> Rick	ECM	0.00
<i>Tricholoma sulphurescens</i> Bres.	ECM	0.03
<i>Tricholoma virgatum</i> (Fr.) P. Kumm.	ECM	0.26
<i>Tricholomella constricta</i> (Fr.) Zerova ex Kalamees	S	0.00
<i>Tricholomopsis rutilans</i> (Schaeff.) Singer	S	0.21
<i>Xerocomellus porosporus</i> (Imler ex Watling) Šutara	ECM	0.03
<i>Xerocomus ferrugineus</i> (Schaeff.) Alessio	S	0.06
<i>Xeromphalina caucinalis</i> (Fr.) Kühner & Maire	S	0.00

productivity were affected by similar weather variables. This similarity may be due to the fact that ECM sporocarps are typically larger than saprotrophic sporocarps (Bässler et al., 2015). Indeed, both total and ECM sporocarp productivity were favoured by August and September accumulated rainfall plus warmer temperatures during the late mushroom season, determining a fruiting period between late summer end and the beginning of winter. This is explained because during these months there is higher water availability in the study area combined with mild temperatures during late summer and the beginning of autumn. Another possible factor propitiating ECM mushroom fruiting,

as Egli (2011) reported, is that the starting point of forest mushroom season is at the beginning of autumn because it coincides with the moment in which the growth of associated trees is completed, and the photosynthesis assimilates are available. Possibly these three factors are key components of ECM fruiting.

For saprotrophic sporocarp productivity, we found a positive effect of September and October accumulated rainfall, as well as a positive effect of higher temperatures at the end of the mushroom season, similar to the effects observed for total and ECM productivity. In this case, we infer that the saprotrophic sporocarps fungal community emerges whenever the environmental conditions are suitable, as they are decomposers and obtain carbon directly from organic matter (Bahram and Netherway, 2022). Moreover, in consonance with our findings, Kaurserud et al. (2012) reported that saprotroph fruiting extends later in autumn than ECM fungi, having a wider fruiting season.

Based on our findings and the reviewed literature, our analysis indicates a high sporocarp richness in *Pinus uncinata* stands. Namely, we recorded 255 species in an area of 1800 m² (18 plots of 10 × 10 m) placed in an altitudinal range from 1648 to 1933 m.a.s.l and surveyed for 5 years (Table 2). Previous research (Alday et al. 2017) accounted for 580 sporocarp species living in *P. sylvestris* stands between 684 and 1.615 m.a.s.l in 1900 m² surveyed for eight years. Salerni et al., (2020), in a *P. nigra* area of 400.000 m² located at 780 – 960 m.a.s.l. and surveyed for 2 years, found a total of 266 species. Finally, Fernandez-Toiran et al. (2006) found 88 species within an area of 3.150 m² of *Pinus pinaster* stands located between 100 and 1200 m.a.s.l. and surveyed for 6 years. Thus, although our study area size is smaller than the compared studies, the order of magnitude of species richness is rather high compared to other *Pinus* stands.

Regarding both sporocarp richness and evenness, we found similar results when comparing them to the ones obtained for ECM sporocarp productivity. A plausible explanation for this similar response to weather variables is that higher productivity values are related to higher richness values as reported for *Pinus sylvestris* of the central Pyrenees (Bonet et al. 2004). This suggests that years with better fruiting conditions would lead to more species fruiting and increasing sporocarp productivity. For the saprotroph community, we did not detect this pattern since higher productivity values occurred in September and higher richness values in October. A plausible explanation for this is that genera producing larger fruit bodies as *Macrolepiota*, *Ampulloclitocybe* or *Rhodocollybia* emerged more during September and reduced their productivity during October resulting in lower productivity and higher species richness in October for the saprotrophic fungal community.

Based on the information mentioned above, it seems that fungal fruiting is induced by late summer precipitation and the prolongation of such fungal fruiting depends on the occurrence of sustained precipitation through the fruiting season. Temperature also plays a crucial role, with mild temperatures during late summer and warmer temperatures during autumn promoting optimal conditions for fungal fruiting. The convergence of these two environmental factors in our study area promotes a high sporocarp fruiting in comparison with forest ecosystems at lower altitudes within Mediterranean areas.

4.2. Fungal community composition

In our study, we observed that ECM sporocarps exhibited higher biomass values while saprotroph sporocarps were represented by lower values, consistent with findings by Alday et al. (2017) in *Pinus sylvestris* stands. Furthermore, when comparing our results with the data recorded in *P. pinaster* in Central Spain and *P. sylvestris* in Central Pyrenees (Fernandez-Toiran et al., 2006; Bonet et al., 2004), we found that *Russula*, *Tricholoma* and *Suillus* were the most abundant genera in our higher productivity families (*Russulaceae*, *Tricholomataceae* and *Boletaceae*), consistent with previous reports. At the species level, we also identified some similarities with other *Pinus* stands. For instance, in *Pinus sylvestris* forests surveyed in the Pyrenean area by Alday et al. (2017), they

identified species such as *Amanita muscaria*, *A. ovoidea*, *A. pantherina*, *Suillus bovinus*, *S. granulatus*, *S. luteus*, *S. variegatus*, *Russula chloroides*, *R. delica*, *R. sanguinea*, *R. torulosa*, and *Tricholoma* species such as *T. imbricatum* and *T. portentosum* while others as *Lactarius deterrimus* seemed to be more linked to *Picea abies* and *Arctostaphylos uva-ursi* (Mühlmann and Göbl, 2006).

Our study revealed that both total and ECM sporocarp community composition responded similarly to certain variables. We observed a shared trend when both communities are significantly related to autumn weather variables that mark the start of fungi fruiting i.e., August accumulated rainfall, and variables that indicate the end of the fruiting season i.e., November mean minimum temperature. This is explained by the fact that total community relative abundances are mainly composed of ECM sporocarps, while saprotroph sporocarps represented a much lower amount of biomass. The year effect was another common significant variable for total and ECM community composition. We could expect this result since ECM fungi fruiting is constricted by the weather conditions in addition to the indirect effects by host plants, both changing interannually. In contrast, we also expected the year effect absence on saprotroph community composition as they grow directly in the upper layers of the soil, decomposing the organic matter and consequently, relying directly on the local site characteristics (Alday et al., 2017) and resulting in a lower seasonal variability as previously described by Santalahti et al. (2016) and Gorfer et al. (2021).

Further investigation of the local site characteristics revealed a significant effect on the sporocarp community composition, consistent with previous studies conducted in high-altitude *Pinus* stands (Martínez-Peña et al., 2012). Thus, we found a significant relation between the organic matter (OM) content and the total community composition in addition to pH for all three studied communities, namely: total, ECM and saprotrophic community composition. The influence of OM content can be explained by the fact that it serves as the primary source of direct energy for saprotrophs (Smith and Read, 2010) and as the secondary source for some ECM species such as *Suillus* (Shah et al., 2016; Colpaert and van Tichelen, 1996). Moreover, OM content improves soil water content (Hudson, 1994), thus benefiting both saprotrophic and ECM species. Moreover, although we found some species that seemed to be related to slightly higher pH values, such as *Tricholoma portentosum* or *Mycena pura*, and other species such as *Boletus edulis* that were related to more acidic soils as demonstrated by Martínez-Peña et al. (2012), the fungal community is found to be growing similarly in all plots since pH values did not differ significantly (Table 1). These results, therefore, need to be interpreted carefully, as the studied plots are located in the same area and their physicochemical soil properties presented a reduced range of values. Besides, it explains the absence of clear differences between species fruiting in different plots, resulting in a random appearance of the sporocarp community composition.

Finally, when comparing our dataset with the IUCN Red List we found that only 6.7% of the species have been assessed, being not representative of the total community of the studied stands. For all least concern species detected the population size is likely to be very large since they are widespread species and without a declining trend. It is worth highlighting that *Entoloma porphyrophaeum*, described as vulnerable and in decline (IUCN, 2022), grows in semi-natural grasslands, and its sporadic appearance in the studied mountain pine stands may be due to its occurrence outside its natural niche. On the other hand, *Gomphidius roseus* and *Imleria badia*, which are categorized as least concern, appeared only one year and only one single occurrence was recorded for *G. roseus*. *G. roseus* is an uncommon but widespread species in Europe and is associated with the presence of *Suillus bovinus* (Olsson et al., 2000). In our case, *S. bovinus* is a very common species that was present every year, indicating that some other unknown factors are related to this fruiting process and further research is needed. *I. badia* is also a widespread species in North America and Europe and there is no indication of decline (IUCN, 2022). With the previous information, we have found that the IUCN assessed species appeared sporadically and

whenever the environmental conditions are adequate. There is a need to investigate the specific requirements for individual fungal species and the conservation status of fungal species at different spatial scales. It is crucial to monitor the species populations regardless of their IUCN category because i) it is a global Red List and some local extinctions may go unnoticed and ii) just because a species is classified as stable it does not mean that it cannot be endangered in the near future, and vice versa. We encourage the creation of specific Red Lists at the national level and/or by specific ecosystems, such as *Pinus uncinata* forests, in order to monitor fungal populations and to have a better perspective on the local to the global conservation status of fungal communities.

5. Conclusions

Our results suggest that the *Pinus uncinata* forests exhibit one of the highest fungal productivity and richness values in the Mediterranean basin, with the higher values recorded during September and October. In addition, it is highlighted the importance of weather conditions in shaping the sporocarp productivity, richness and together with the soil properties when affecting the sporocarp community composition of this mountain pine ecosystem. Our findings also highlight the dominance of ECM species over saprotrophic species in these forests in both productivity and species richness. Moreover, we have reported the appearance of certain species that are commonly reported in different pine ecosystems at lower altitudes. Finally, the presence of certain species listed in the IUCN Red List emphasizes the necessity for additional research on the conservation of the fungal community in these forests. This can be achieved by integrating traditional sporocarp sampling within molecular techniques and conducting simultaneous monitoring of the species in other comparable areas. Overall, our research significantly contributes to the knowledge of fungal ecology in *Pinus uncinata* forests and highlights the importance of deeper investigations into fungal dynamics within forest ecosystems, emphasizing the need for comprehensive research efforts in these ecologically valuable areas.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

Appendix A.

Fig. A1

Table A1

Appendix B.

Table B1

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